

BALASUBRAMANYA N

SOFTWARE ENGINEER

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Summary

Backend-focused Full Stack Developer with experience building scalable systems in utilities and healthcare domains using .NET, Spring Boot, and Angular. Skilled in designing distributed backend services, billing workflows, and microservices with complex business logic. Experienced in system design, API design, and secure authentication and authorization (RBAC, MFA), with a strong focus on performance, data consistency, and production-grade reliability.

Experience

Esyasoft

Software Engineer

Feb 2025 – Present

Tools Used: .NET, React, Entity Framework Core

- Designed and developed a scalable revenue management platform for electricity, water, and gas utilities, supporting end-to-end workflows from consumer onboarding to billing and payment processing.
- Built high-throughput billing pipelines to process and generate bills for lakhs of consumers, including bulk ingestion of meter readings via CSV and automated monthly billing cycles.
- Engineered backend services to handle complex business logic involving tariffs, taxation, advance payments, rebates, and due calculations while ensuring data consistency and accuracy.
- Optimized data processing for large-scale workloads, enabling efficient handling of high-volume transactions with minimal performance degradation.
- Improved system flexibility by leveraging Entity Framework to support multiple client databases without requiring code changes.
- Developed multilingual support and modular architecture to enable adaptability across different regions and clients.

Prodat IT Solutions

Software Associate

Feb 2023 – Feb 2025

Tools Used: Spring Boot, Angular, Hibernate

- Designed and developed a microservices-based healthcare platform encompassing modules for patient management, claims processing, insurance handling, and appointment scheduling.
- Integrated OCR-based automation pipelines leveraging Azure and AWS cloud services to streamline document processing and reduce manual data entry.
- Ensured full compliance with HIPAA regulations, standards, and best practices to maintain data privacy and security across all healthcare modules.
- Optimized database queries and backend logic to enhance system performance, reliability, and response times under high-load conditions.

Technical Skills

Languages: C#, Java, JavaScript, TypeScript

Frontend: Angular, React, HTML, CSS, Bootstrap, Angular Material

Backend & Frameworks: .NET, Spring Boot, WebSocket, Hibernate, Entity Framework Core

Databases: PostgreSQL, MySQL

Cloud & DevOps: Azure, AWS, Docker, Git, Azure DevOps

Microsoft Power Platform: Power Apps, Power Pages, Power Automate, Dataverse

Tools: Jira, Postman

Projects

Document Me – Real-time Collaborative Editor 🔄

- **Tech Stack:** Angular, ASP.NET Core, Yjs, PostgreSQL, WebSocket
- Implemented CRDT-based conflict resolution using Yjs and Hocuspocus WebSocket server to ensure consistent document state across all connected clients.
- Secured the platform using JWT authentication with HttpOnly cookies, middleware-protected REST APIs, and WebSocket-level connection validation.

Am I Authorised – Role-Based Access Control 🔄

- **Tech Stack:** Angular, ASP.NET Core, PostgreSQL, Bootstrap, Angular Material

- Built a full-stack RBAC application enforcing fine-grained, backend-driven authorization across pages, menus, buttons, and actions based on user roles and permissions.
- Designed a JWT-based auth system embedding both role and permission claims, ensuring every API request is validated server-side before processing.
- Implemented four distinct roles — Admin, Auditor, Employee, and Manager — each with specific, declarative permissions extensible for future requirements.

Double Trouble – Multi-Factor Authentication 🔒

- **Tech Stack:** Spring Boot, Thymeleaf, PostgreSQL, Spring Security, HTTP Session
- Built a two-step MFA system using Spring Boot and Spring Security, enforcing both password and TOTP verification before granting access.
- Integrated QR code-based TOTP setup using the `dev.samstevens.totp` library, compatible with standard authenticator apps like Google Authenticator.
- Leveraged Spring Security's built-in MFA support over custom filter logic, resulting in a cleaner, maintainable, and more secure implementation.

Education

Jain (Deemed-to-be University)

Master of Computer Applications – Computer Science

Bangalore, India

Jan 2028

Certifications

Java Course – Mastering the Fundamentals | *Scaler*

Foundational C# with Microsoft | *freeCodeCamp × Microsoft*

Angular Foundations Course | *Scholar Hat*